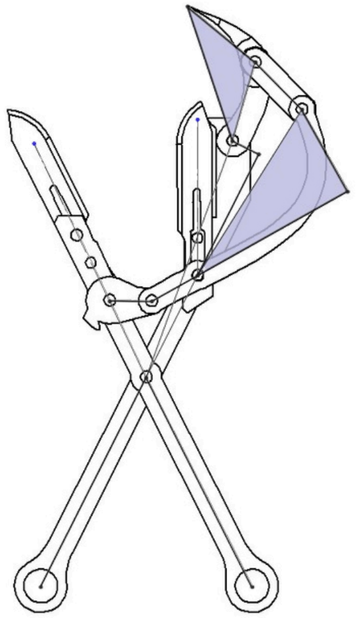
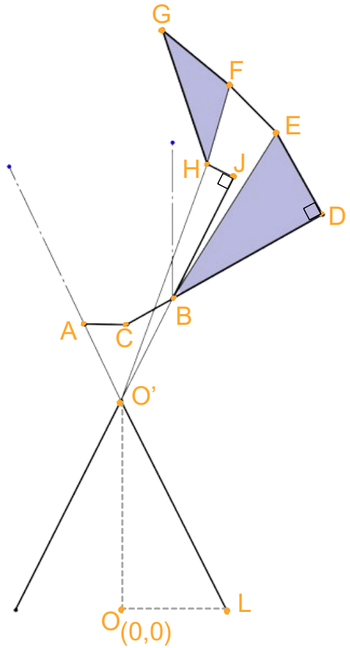
PROJECT 1

KINEMATIC PEDICEL CUTTER

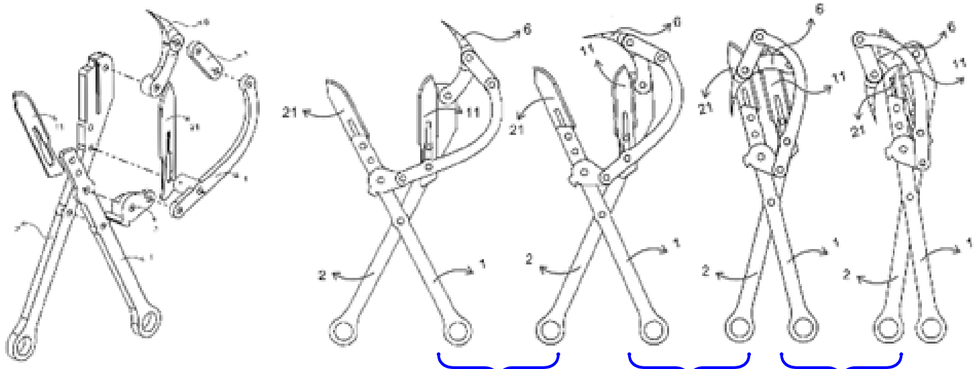




KINEMATIC PEDICEL CUTTER

BRIEF BACKGROUND

During the autonomous cutting of tomatoes we often face the problem with the cutting end effector. These systems usually rely on point cloud visual data via AI and computer vision to identify the cutting point. These points often have discrepancies, leading to inaccuracies in the exact position of the cutting point. This can result in incomplete cuts or damage to the fruit itself.
Thus we need to have a hook mechanism which pulls out the pedicel from a zone of possibilities of occurrence as per the detected cutting point. Existing tools and automated systems also often lack the capability to securely pull and stabilize the pedicel during cutting, which is crucial for ensuring a clean and precise cut.



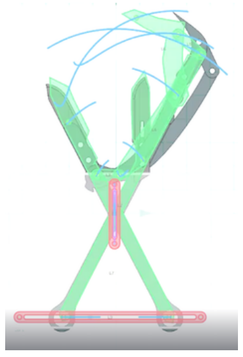
PHASES OF THE MOTION OF FINAL DESIGN

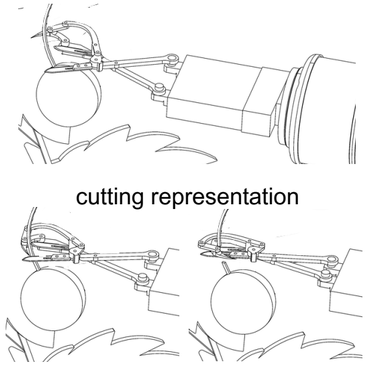
IDEATION AND PROTOTYPES

Started with a 8-bar kinematic cutter with integrated hook and pull mechanism to compensate for discrepancies of point cloud visual data. Optimised the design after 20+ prototypes for promising result. The linear translation of main arms (1 & 2) results the phases such as hooking phase, pulling phase and holding/cutting phase.



VARIOUS DIFFERENT PROTOTYPES TO OPTIMISE THE CUTTING AREA, STRENGTH AND PRECISION









DERIVATION OF GOVERNING EQUATION

DERIVED THE HOOK TRAJECTORY BY DEFINING 40 DIFFERENT PARAMETERS (VARIABLES & CONSTANTS) AND SOLVING 20 LINEAR/TRIGONOMETRIC EQUATIONS

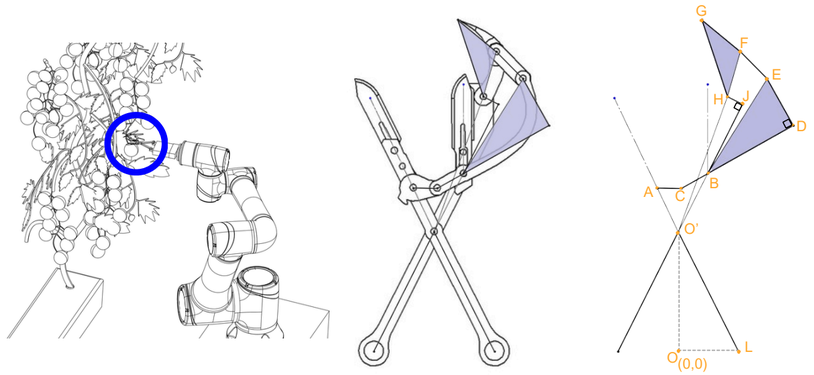


Fig 1. Kinematic representation of the mechanism

DEFINING OF PARAMETERS OF THE MECHANISM

For the derivation of the governing equation for the trajectory of the hook (x,y) with respect to input angle θ and scissor constants, we define parameters such as -

Input angle :- θ, Output angle :- α, Trajectory points :- (x,y) [To find] and (x',y')
Scissor constants (Does not change in operation) :- Lengths - l_1, l_2, l_3, d, a_1, a_2, a_3, a_4, a_5, a_6, a_7
Scissor constants (Does not change in operation) :- Angles - λ, ψ_1, ψ_2, ψ_3
Variables during operation :- lengths - b_1, b_2, b_3
Variables during operation :- angles - Φ_1, Φ_2, Φ_3, Φ_4, Φ_5, Φ_6, Φ_7, γ_1, γ_2, γ_3, γ_4, γ_5
Origin :- O, End points :- O', A, B, C, D, E, F, G, H, J, K

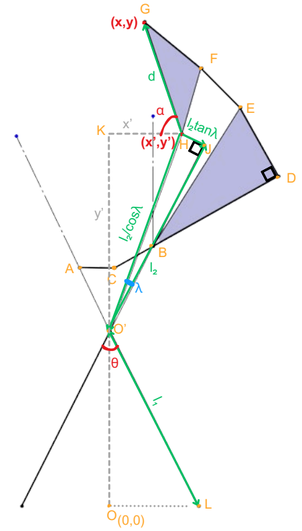


Fig 2. Determination of trajectory of hook (x,y) with respect to α and (x',y')

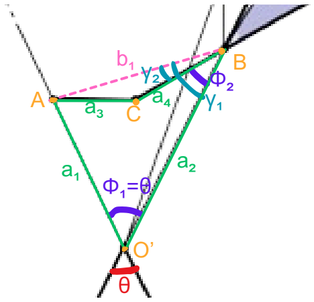


Fig 3. Mid section of the mechanism

According to *Fig. 2,* In △*GHK*
$x = x' - d\cos\alpha$
$y = y' + d\sin\alpha$ } *eq. 1*
Now, *KH* = x', *OK* = y', ∠*KO'H* = θ/2 - λ
$\Rightarrow x' = \frac{l_2 \sin(\theta/2 - \lambda)}{\cos\lambda}$ & $y' = l_1\cos(\theta/2) + \frac{l_2\cos(\theta/2 - \lambda)}{\cos\lambda}$ } *eq. 2*

By *eq.1* and *eq. 2* we get :-
$x = \frac{l_2 \sin(\theta/2 - \lambda)}{\cos\lambda} - d\cos\alpha$ & $y = l_1\cos(\theta/2) + \frac{l_2\cos(\theta/2 - \lambda)}{\cos\lambda} + d\sin\alpha$ } *eq. 3*

From *Fig. 3*, $\Phi_1 = \theta$
In △*AO'B,* by cosine law -
$b_1 = \sqrt{a_1^2 + a_2^2 - 2a_1a_2\cos\theta}$ } *eq. 4*
By sine law -
$a_1/\sin\gamma_1 = b_1/\sin\theta$
$\Rightarrow \gamma_1 = \sin^{-1}(a_1\sin\theta/b_1)$ } *eq. 5*

In △*ABC*, by cosine law -
$\gamma_2 = \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right)$ } *eq. 6*

∵ $\Phi_2 = \gamma_1 - \gamma_2$, ∴ by *eq. 4, 5* & *6*
$\Phi_2 = \sin^{-1}(a_1\sin\theta/b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right)$ } *eq. 7*

where, $b_1 = \sqrt{a_1^2 + a_2^2 - 2a_1a_2\cos\theta}$

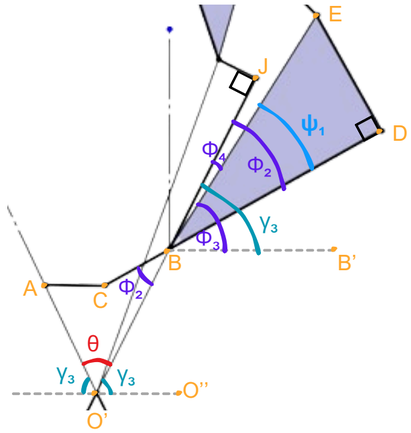


Fig 4. Right portion of the mechanism

From *Fig. 4*,
$\gamma_3 = \pi/2 - \theta/2$ } *eq. 8*
$\Phi_4 = \Phi_2 - \psi_1$ } *eq. 9*
From *eq. 9* and *eq. 7*
$\Phi_4 = \sin^{-1}(a_1\sin\theta/b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right) - \psi_1$ } *eq. 10*
$\Phi_3 = \gamma_3 - \Phi_4$; thus by *eq. 8* and *eq. 10*
$\Rightarrow \Phi_3 = \pi/2 - \theta/2 + \psi_1 - \sin^{-1}(a_1\sin\theta/b_1) + \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right)$ } *eq. 11*

CONTINUATION ->

DERIVATION CONTINUATION AND RESULT

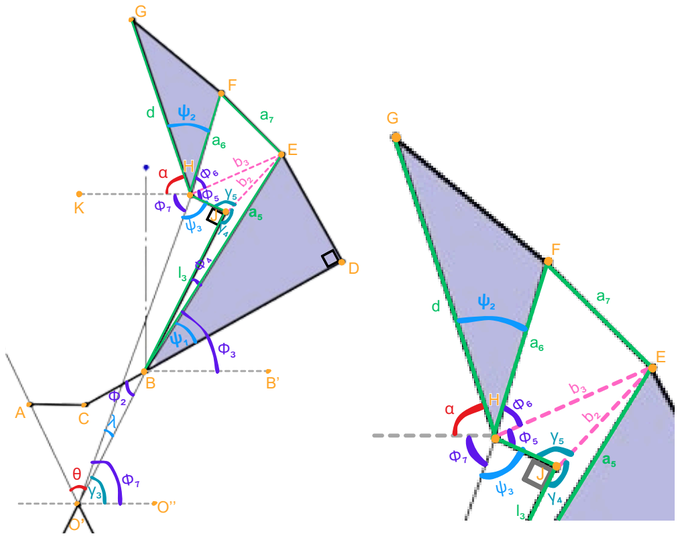


Fig 5. Determination of trajectory of hook (x,y) with respect to θ and scissor constants

From Fig. 5,
$\Phi_7 = \gamma_3 + \lambda$; thus by eq. 8
$\Phi_7 = \pi/2 - \theta/2 + \lambda$ } eq. 12

Around point H,
$2\pi = \alpha + \psi_2 + \Phi_5 + \Phi_6 + \Phi_7 + \psi_3$ =>
$\alpha = 2\pi - (\psi_2 + \psi_3 + \Phi_5 + \Phi_6 + \Phi_7)$ } eq. 13

In △HJO',
$\psi_3 = \pi/2 - \lambda$ } eq. 14

In △BJE, by cosine rule,
$b_2 = \sqrt{a_5^2 + l_3^2 - 2a_5 l_3 \cos\Phi_4}$ } eq. 15
by sine rule,
$a_5/\sin(\pi-\gamma_4) = b_2/\sin\Phi_4$ =>
$\gamma_4 = \pi - \sin^{-1}(a_5 \sin\Phi_4 / b_2)$ } eq. 16

Around point *J*,
$2\pi = \gamma_4 + \gamma_5 + \pi/2$ => by *eq. 16*
$\gamma_5 = \pi/2 + \sin^{-1}(a_5 \sin\Phi_4 / b_2)$ } *eq. 17*

In △*HJE*
$HJ = l_2 \tan\lambda$ *(from Fig.2)* ,
by cosine law
$b_3 = \sqrt{(l_2 \tan\lambda)^2 + b_2^2 - 2 l_2 \tan\lambda \, b_2 \cos\gamma_5}$ } *eq. 18*
Now by cosine law again,
$\Phi_5 = \cos^{-1}\left(\frac{(l_2 \tan\lambda)^2 + b_3^2 - b_2^2}{2 l_2 \tan\lambda \, b_3}\right)$ } *eq. 19*

In △*HFE*, by cosine rule
$\Phi_6 = \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2 a_6 b_3}\right)$ } *eq. 20*

∴ In *eq. 13*,
$\alpha = 2\pi - \psi_2 - \psi_3 - \Phi_5 - \Phi_6 - \Phi_7$ put *eq. 12, 14, 19, 20* =>
$\alpha = 2\pi - \psi_2 - \cos^{-1}\left(\frac{(l_2 \tan\lambda)^2 + b_3^2 - b_2^2}{2 l_2 \tan\lambda \, b_3}\right) - \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2 a_6 b_3}\right) - \pi/2 + \theta/2 - \lambda - \pi/2 + \lambda$

FINAL GOVERNING EQUATION -

$$x = \frac{l_2 \sin(\theta/2 - \lambda)}{\cos\lambda} - d\cos\alpha \quad \& \quad y = l_1 \cos(\theta/2) + \frac{l_2 \cos(\theta/2 - \lambda)}{\cos\lambda} + d\sin\alpha$$

where by eq. 4,15,17,18,10 :-

$$\alpha = \pi - \psi_2 + \theta/2 + \cos^{-1}\left(\frac{(l_2 \tan\lambda)^2 + b_3^2 - b_2^2}{2 l_2 \tan\lambda \, b_3}\right) - \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2 a_6 b_3}\right)$$

$$b_3 = \sqrt{(l_2 \tan\lambda)^2 + b_2^2 - 2 l_2 \tan\lambda \, b_2 \cos\gamma_5}$$

$$b_2 = \sqrt{a_5^2 + l_3^2 - 2 a_5 l_3 \cos\Phi_4}$$

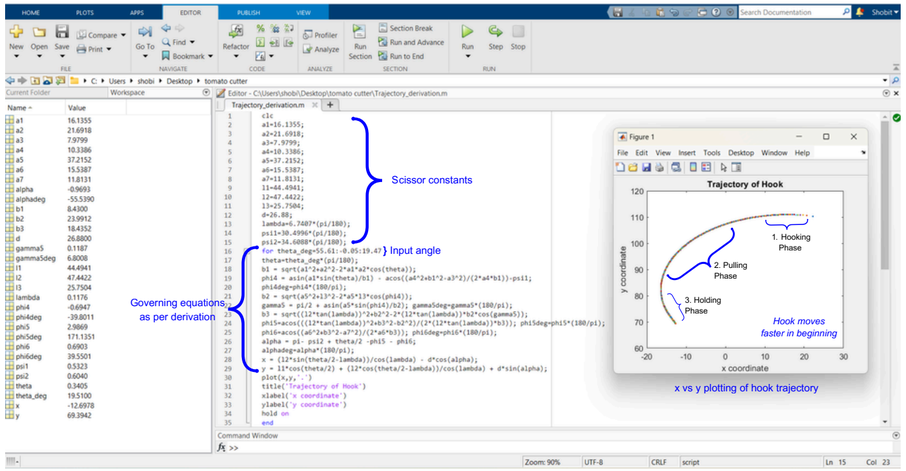
$$\gamma_5 = \pi/2 + \sin^{-1}(a_5 \sin\Phi_4 / b_2)$$

$$\Phi_4 = \sin^{-1}(a_1 \sin\theta / b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2 a_4 b_1}\right) - \psi_1$$

$$b_1 = \sqrt{a_1^2 + a_2^2 - 2 a_1 a_2 \cos\theta}$$

EXECUTION OF THE THEOREM IN MATLAB

θ *(input angle)* ∈ **[19.47°, 55.61°]** {*Kinematically constraint*}



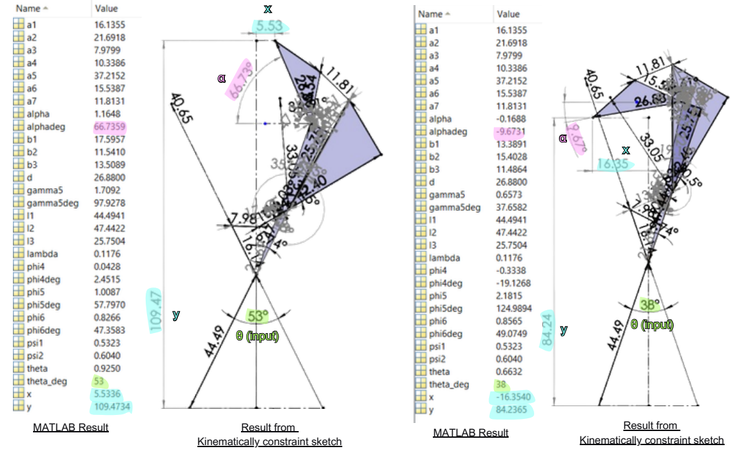


Fig 6. Exact matching of values of the MATLAB Result and the result obtained from Kinematically constraint Sketch in SolidWorks

RESULT :- **GOVERNING EQUATION IS CORRECT, THUS CUTTING AREA AND HARVESTING PRECISION CAN BE OPTIMISED MATHEMATICALLY**